L8.1: Orthgonality and linear independence

Transcript Language

English

Hello, and welcome to the Maths 2 component of the online B.Sc. program in data science.

In this video, we are going to talk about orthogonality and linear independence.

So, we have seen before the notion of an inner product

and in this video, we are going to continue using the ideas of inner products and norms.

So, let us recall first that we have expressed the angle between two vectors in terms of the special inner product, namely the dot product in Rn, and the special norm, namely the length in Rn. So,

how do we do that? We take the angle theta between two vectors, let us say,

the vectors u and v. And of course, when we talk about angle, we first,

we have to first look at the plane that they span and then we look at the angle on that plane.

And how do we compute it? To compute it, we can use the dot product or the norm and the norm,

namely, cosine of theta is equal to u dot v divided by norm of u times norm of v.

So, the dot product is a special case of an inner product on Rn and the length

is a special case of a norm on Rn. So, let us first ask what is the geometric

intuition of what are called orthogonal vectors. We want to study orthogonality in this video.

So, what is a pair of orthogonal vectors? So, the angle theta between two vectors u and v in Rn,

if it is a right angle, which means it is 90 degrees, then cosine of theta is 0.

And we can, we have just seen in the previous slide that we can compute it by looking at

the dot product of these two vectors divided by the product of the norms. So, that means,

0 is u dot v divided by norm of u times norm of v. Of course, if we have a, if we have a ratio,

which is 0, then the, that means the numerator is 0. So, this means u dot v is 0.

So, this is exactly what we mean by saying two vectors are orthogonal.

So, two vectors are orthogonal when in Rn when they form a right angle, they

are at right angles to each other. So, in R2, for example, this is what we mean by two vectors are

orthogonal. So, this is a geometric intuition. Let us, as an example, let us look at the vectors

1, 2, 3 and 2, 2, minus 2. They are orthogonal, because if you compute the dot product,

then you will see this is 1 times 2 plus 2 times 2 plus 3 times minus 2, which is 2 plus 4 minus 6,

which is 0. So, that means cosine of theta is 0. And cosine of theta is 0 means you can back calculate to say that theta is 90 degrees. So, let us define the notion of an orthogonal

vectors in the general vector space v. And the intuition we have just seen is

that if the dot product is 0, then in Rn the vectors are said to be orthogonal. So, we can

make a more general definition now. So, suppose you have an inner product space v and you have

two vectors u and v then we say that they are orthogonal if you compute the inner product u,v

and that is 0. So, the, let us do an example. Suppose you take R2 with the inner product

given by this complicated looking function, which we have seen in the previous video. So,

you have the inner product of u,v is x1y1 minus x1y2 minus x2y1 plus 2 times x2y2, and u is x1x2,

v is y1y2. So, then the vectors 1, 1 and 1, 0 are orthogonal in this inner product.

Let us compute why that is the case. So, if you compute it for this inner product, what we get,

so we get 1 times 1 minus x1 times y2, 1 times 0 plus x2 times y1, 1 times 1 plus 2 times x2 times y2, which is 1 times 0. And you can see this is 1 minus 0 minus 1 plus 0, so this is 0.

So, the inner product is 0 and that is why these vectors are orthogonal by definition.

I want to point out that in this case we chose this particular inner product

on R2. Had we chosen the usual inner product namely the dot product, then these vectors are

not orthogonal. So, the notion of orthogonality depends on the chosen inner product, even for Rn,

it depends on which product to choose. Now, if we say without alluding to any inner

product that the vectors are orthogonal then by convention it means that we are talking about the

usual inner product namely the dot product. But if we specify some other inner product

and talk about orthogonality, it means with respect to that inner product.

So, let us talk about an orthogonal set of vectors. So, an orthogonal set of vectors in

an inner product space V is a set of vectors whose elements are mutually orthogonal, that means they are pair-wise orthogonal. So, if you take any two of them, then they are orthogonal. So, explicitly

what that means is if you take the set v1, v2, vk in V then S is an orthogonal set of vectors.

If you look at vi,vj that is 0, of course, you need that i is not equal to J. So,

if you take v1 and v2 are orthogonal v1 and v3 are orthogonal, v1 and v4 are

orthogonal, all the way up to v1 and vk, similarly v2 and v3, v2 and v4,

all the way up to vk and so on, that is what we mean by mutually orthogonal.

For example, let us consider R3 with the usual inner product, the dot product, then the set S,

which is 4, 3, minus 2, minus 3, 2, minus 3 and minus 5, 18, 17 is an orthogonal set of

vectors. Let us quickly see why that is the case. So, let us compute these dot products.

So, if you dot this with minus 3, 2, minus 3, then you get 4 times minus 3 plus

3 times 2 plus minus 2 times minus 3 which is minus 12 plus 6 plus 6 which is 0.

So, certainly the first two are orthogonal. Let us look at the next, maybe the first and the third.

So, if you do this, you get minus 20 plus

54 minus 34 and indeed that 0. And finally, if you do the second and the third one,

then we get minus 15 plus 36 minus 51 and oh, I made a mistake here,

sorry, this is plus 15, plus 15 plus 36 minus 51 which is 0. So, these are indeed

mutually orthogonal. So, this is an orthogonal set of vectors.

So, this is, now let us talk about orthogonality and linear independence. This is the title of our

video and clearly this is the most important slide in this video. So, the reason we want to discuss

orthogonality, is one of the reasons at least, is because we want to, because it has some relation

with when vectors are linearly independent. So, let us see what we want to say.

So, suppose we have a orthogonal set of vectors in the inner product space V, then

v1, v2, vk is a linearly independent set of vectors. So, if you have an orthogonal set, automatically it is a linearly independent set. Now, let us recall, to check something

is linearly independent was a fairly hard process. We had to write down summation civi

is 0 and then by some way we had to say that each of the coefficients Ci is 0.

And in Rn if we had to do this, that means we had to solve a system of linear equations

and check that indeed each coefficient is ending up being 0. So, that was a fairly non-trivial

process. Instead, here what we are saying is, if it so happens that these v1, v2, vk are

an orthogonal set of vectors and notice that to check something is orthogonal is quite easy.

You have to just check that the inner products are 0, which is a fairly mechanical process.

Let us see a quick proof or at least idea of the proof of this statement

why is this happening. So, suppose, maybe I will do this in red. So, suppose summation civi is 0,

well, then what I can do is I can take the inner product of summation civi

with vi or maybe let us say with v1, let us do it for v1, and you will see the same thing happens for any vi.

So, if you do that, then this is the inner product of 0 with v1 and the inner product of 0

with anything is 0. This is one of the things that we know about inner products. You can check this

from the definition, because you can write 0,v1, 0,v1 plus 0,v1 and then cancel 0,v1 on both sides.

So, this is 0. But on the other end we know what this is,

the first term is. So, this is a sum. So, I can take out my summation.

So, therefore, summation i is equal to 1 through k, ci times vi,v1 is 0.

Therefore, summation i is 1 through k, I can take out the constant also, vi,v1 is 0 and therefore,

ci. So, what are the non-zero terms here? Remember, that these are an orthogonal set of

vectors. So, unless i is equal to 1, everything is 0, because vi,v1 is equal to inner product of

v1,vi. So, inner product of vi and v1 is 0 for all i not equal to 1. So, the only term that I

am left with here is c1 times v1,v1 and this is 0. And so, if v1 is not 0, then this inner product is non-zero, so and that will imply that c1 is 0. So, maybe here I should make a

small correction to what I have written here. Namely, I should say, let v1, v2,

vk be an orthogonal set of non-zero vectors. So, typically, when we say orthogonal set,

we often mean non-zero vectors. But that is not, that is not the definition per set. So,

we will say orthogonal set of non-zero vectors. So, if you have an orthogonal set of non-zero

vectors, then this part v1, v1 is not 0 and as a result c1 is equal to 0.

And you can do this argument for instead of v1 you could have done it with v2

or v3 or vj in general and you get that cj is 0. So, that will tell you that each of these

constants is 0, which is exactly the definition of linear independence. So, that is the reason

why every orthogonal set of non-zero vectors is indeed a linearly independent set of vectors.

So, what is an orthogonal basis? So, let V be an inner product space. A basis consisting of mutually orthogonal vectors is called an orthogonal basis. So, this is the same as saying

that an orthogonal basis is an

orthogonal set which is also a basis or a basis which is also an orthogonal set.

So, since an orthogonal set we saw is linearly independent, an orthogonal set is a basis exactly

when it is a maximal orthogonal set. Because how do we, when is a linearly independent set a basis?

When it is the maximal linearly independent set. So, if it is already orthogonal, then that is

the same as saying it is a maximal orthogonal set or maximal linearly independent set.

So, what does maximal orthogonal set mean? It means there is no orthogonal set which

contains strictly contains this set that is being referred to. So, a set is called

a maximal orthogonal set if there is no super set which is also an orthogonal set.

So, specifically, what this means is if the dimension of V is n, then an orthogonal basis

is exactly an orthogonal set of n vectors. So, this is what you should keep in mind.

So, let us look at some examples of orthogonal basis. The standard basis that we have seen is

an orthogonal basis. Of course, I have not referred to an inner product here.

So, as I mentioned before, the convention means that is that if I do not refer to

any particular inner product, you have to take the dot product. So, the standard basis in the

inner product space Rn with the dot product. Let us look at this example which we have done

before. So, we have seen already that this is a orthogonal set.

What is the size of the set? It is a set of size 3. And what is the dimension of R3? It is 3. So,

this is a orthogonal set with three vectors, where 3 is the dimension of the vector space R3. So,

this is going to be an orthogonal basis. We are using this thing here, the statement here.

And finally, let us look at R2 with the inner product that we, the non-standard inner product

that we defined earlier, then we saw that this is a orthogonal set of vectors, and it has size 2.

So, there are two vectors and the dimension of R2 is 2. So, this must be an orthogonal basis. So, this is also an orthogonal. Again, we are using this fact that is written here.

So, let us recall what we saw in this video. So, we have seen the notion of an orthogonal,

when two vectors are orthogonal, so a pair of orthogonal vectors or we can just say just

orthogonal vectors, remember that this depends on the inner product chosen and the intuition

is coming from the fact that in R2 or in Rn, if, with respect to the usual dot product, if

vectors are orthogonal, then that exactly means that they are at a right angle to each other,

like this perpendicular to each other. So, in general, we can talk about the notion

of orthogonal vectors for any inner product space. And

this is a very useful notion, because if you have an orthogonal set, which means

all the vectors are mutually orthogonal, then that set is actually linearly independent. So, now,

we can check linear independence using the notion of orthogonality, this is the main point.

And after that we defined the notion of an orthogonal basis, namely its orthogonal set which is also a basis and if you have your vector space as dimension V,

essentially what this means is that you have a set of n or n vectors which formed

an orthogonal set and we have seen the examples out here. Thank you.